

From Raw Curves to Harmonic Blueprints: A Conceptual Guide to TE Transformation

In precision engineering and machine learning, we often choose between analyzing granular details or the overarching "rhythm" of a system. In this repository, that choice is codified into two distinct development paths. While our standard models predict Transmission Error (TE) point-by-point, we have introduced a dedicated **Harmonic-Wise Pipeline** to replicate high-level research standards. This path, known as ^{RCIM-MBR}, moves away from predicting individual data points and instead learns to predict the mathematical "blueprint" of a complete TE curve.

1. The Paradigm Shift: Direct vs. Harmonic-Wise Learning

To understand the architecture of this repository, we must distinguish between our two comparison tracks. ^{TE-CVP} **(Direct-TE)** is our standard comparison branch where models predict the TE value for a specific angular position. ^{RCIM-MBR} **(Paper-Faithful)** is our reproduction branch, designed to mirror the logic found in the reference RCIM paper. The following table summarizes the fundamental shift in how the machine learning problem is framed:

Feature	^{TE-CVP} : Direct-TE (Repository Standard)	^{RCIM-MBR} : Harmonic-Wise (Paper-Faithful)
Model Inputs	Angular Position + Operating Conditions	Operating Conditions only (Speed, Torque, Temperature, Direction)
Model Targets	A single TE value for a specific angle	A "stack" of harmonic coefficients for the entire curve
Core Learning Goal	Predict the exact value at a specific dot.	Predict the "recipe" or summary of the curve shape.

By shifting from point-by-point prediction to curve-summary prediction, we align our repository with the scientific logic of advanced research. This approach treats the TE curve as a structured signal rather than a collection of disconnected measurements, allowing for a more compact and physically meaningful representation. This shift in focus requires a specific mathematical mechanism to translate curves into numbers the model can understand.

2. Breaking the Wave: The Anatomy of Harmonic Decomposition

To summarize a curve, we employ **Harmonic Decomposition**. Instead of storing thousands of individual points, we describe the TE curve using a truncated Fourier series:

$$TE(\theta) \approx a_0 + \sum c_k \cos(k\theta) + s_k \sin(k\theta)$$

As an educational architect, it is helpful to view this "recipe" through its constituent ingredients:

- **The Offset (a_0)**: The average baseline of the signal, representing the vertical center of the curve.
- **The Coefficients (c_k and s_k)**: These are the "volume knobs" for the signal. While the repository predicts these as sine/cosine pairs for numerical stability in regression, they map directly to the paper's focus on **Amplitude (A_k)** and **Phase (ϕ_k)**.
- *Technical Note*: $A_k = \sqrt{c_k^2 + s_k^2}$ and $\phi_k = \operatorname{atan2}(-s_k, c_k)$.
- **The Angular Position (θ)**: The independent variable representing the rotation cycle.

- **The Summation ($\sum$):** The process of layering specific frequencies (harmonics) to reconstruct the complex geometry of the original measurement. In this framework, the curve is no longer a series of dots; it is a signal to be summarized. Once translated into these mathematical ingredients, the role of the machine learning model changes fundamentally.

3. Why the Angle Disappears: Operating Conditions as the New Drivers

The most significant conceptual hurdle in the harmonic-wise pipeline is that the **angular position disappears** from the model's input. In a direct model, the angle is a required coordinate. In the harmonic-wise branch, the angle moves from being an input (independent variable) to part of the **output structure** (the reconstructed grid). The model no longer asks "where are we on the curve?" Instead, it asks "what does the environment look

like?" **!IMPORTANT The Input Shift**

- **Before (TE-CVP):** Speed + Torque + Temp + **Angle** $\rightarrow$ One TE Point
- **After (RCIM-MBR):** Speed + Torque + Temp + **Direction** $\rightarrow$ **The Entire Curve Shape** (Coefficients) By including **Direction** as a primary driver alongside Speed, Torque, and Temperature, the model learns how these physical factors dictate the *shape* of the curve. The model predicts the "recipe" for the whole rotation, effectively asserting that at a given set of conditions, the resulting error curve will always follow a specific harmonic pattern. To ensure this pattern is manageable, we must decide which "ingredients" are most important.

4. The Selected Harmonic Palette

We do not need an infinite number of frequencies to describe a TE curve accurately. The pipeline utilizes a "compact representation" by selecting 10 specific harmonics that capture the system's practical periodic structure. These are categorized by their physical significance:

- **Primary Structure:** 0, 1, 3
- **Mechanical Interactions:** 39, 40, 78, 81
- **High-Frequency Details:** 156, 162, 240 Truncating the signal to these specific harmonics allows for a data format that is efficient for computation while remaining "expressive enough" to reconstruct the original curve. However, predicting these coefficients is only half the process; we must eventually turn these abstract numbers back into something we can evaluate against real-world data.

5. From Coefficients to Reconstruction: Closing the Loop

In the repository, the prediction task is handled by `run_harmonic_wise_comparison_pipeline.py`, which utilizes a family of **HistGradientBoosting** regressors—one for each harmonic target. Once these models output the predicted c_k and s_k values, we enter **Stage 6: Reconstruction**. This is the bridge back to physical reality.

1. **Inputting Operating Conditions:** The model is fed the current environmental data (Speed, Torque, Temp, Direction).
2. **Predicting Ingredients:** The regressors output the predicted coefficients for the harmonic palette.
3. **Applying the Formula:** The pipeline uses the harmonic decomposition formula to calculate TE values across a standardized 360-degree angular grid.

4. **Generating a Human-Readable Curve:** This results in a continuous TE curve that can be compared directly against high-fidelity experimental measurements. This reconstruction allows us to see how close our predictions come to the high standards set by the original research paper.

6. Measuring Success: The Path to Target A

Our current progress is measured against **Target A**, which is defined as the offline paper-comparable threshold. This is distinct from **Target B**, which involves future work on real-time online compensation. "The current repository baseline reports a **9.403%** mean percentage error on the test split, while the paper-aligned offline target is **4.7%**." While there is a gap to close, our architectural analysis provides a vital insight: the **"Truncation-only Oracle."** When we bypass the machine learning model and use the actual harmonic coefficients from the data, the error drops significantly below the target. This proves that our **harmonic set is expressive enough** to describe the system perfectly. The bottleneck is not the harmonic method itself, but the **predictor accuracy** of the regressors. This directs our future engineering focus toward optimizing the model's mapping of operating conditions to coefficients.

7. Summary & Key Takeaways

The Harmonic-Wise Pipeline provides the repository with a rigorous, paper-faithful validation path. To master this logic, keep these three **Golden Rules** in mind:

1. **Predict the Shape, Not the Point:** The model focuses on a compact harmonic summary of the entire curve rather than individual data samples.
2. **Angle-Free Inputs:** During training, operating conditions (including Direction) are the only drivers; the angular position is reserved for the final reconstruction grid.
3. **Offline-Only Reproduction:** This pipeline is currently a ^{RCIM-MBR} validation tool. It is designed for faithful comparison with scientific benchmarks, separate from the real-time online compensation loops used in deployment. By utilizing this harmonic blueprint, we ensure our work is not just predictive, but deeply aligned with the mathematical and physical standards of modern engineering research.